

**Read a string and replace lower / upper / digits / spaces
/ special char with @ / # / \$ / * / =**

```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i; clrscr();
puts("Enter the string ");gets(s);
for(i=0;s[i];i++)
{
if(s[i]>='a'&& s[i]<='z')s[i]='@';
else if(s[i]>='A'&& s[i]<='Z')s[i]='#';
else if(s[i]>='0'&& s[i]<='9')s[i]='$';
else if(s[i]==' ')s[i]='*';
else s[i]='=';
}
printf("Converted string is %s",s);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Google Chrome

04:26 PM
05-Apr-25

```
TC
Enter the string
James Bond - 007
Converted string is #@@@@*#@@@@*=$$$$
```

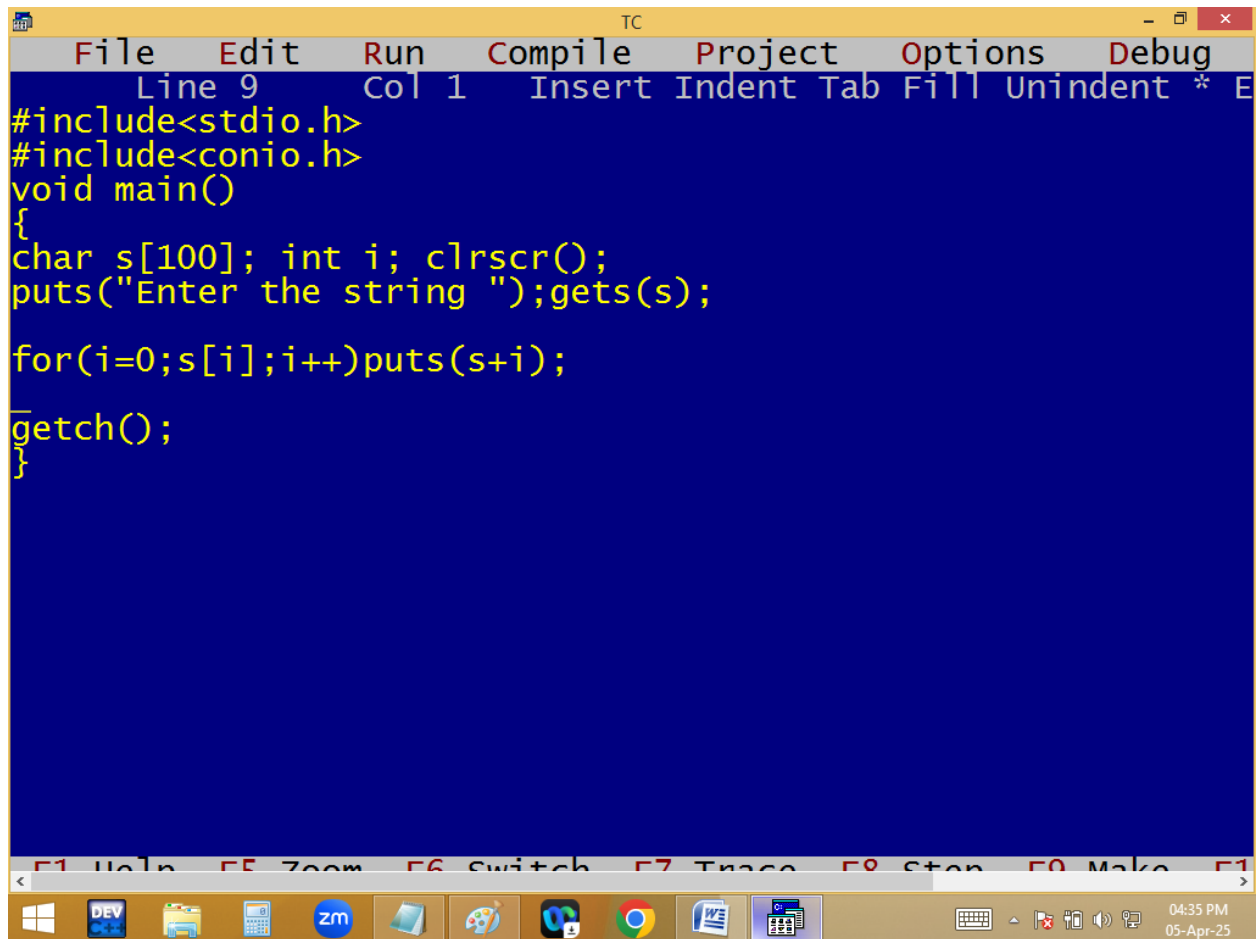
Alia

Alia

lia

ia

a



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 9", "Col 1", and "Insert Indent Tab Fill Unindent * E". The code editor has a blue background and contains the following C program:

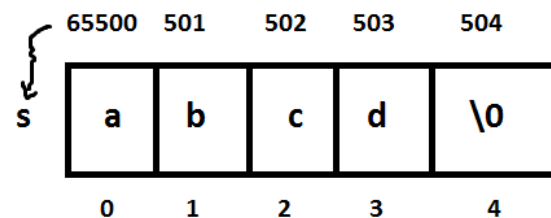
```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i; clrscr();
puts("Enter the string ");gets(s);

for(i=0;s[i];i++)puts(s+i);

getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "04:35 PM" and "05-Apr-25".

```
Enter the string
Bharathi
Bharathi
harathi
arathi
rathi
athi
thi
hi
i
```



```
for( i=0 ; s[i]!='\0'; i++ ) puts( s+i );
```

/ \

$65500 + 0 * 1 = 65500$ to $\backslash 0 \implies abcd$

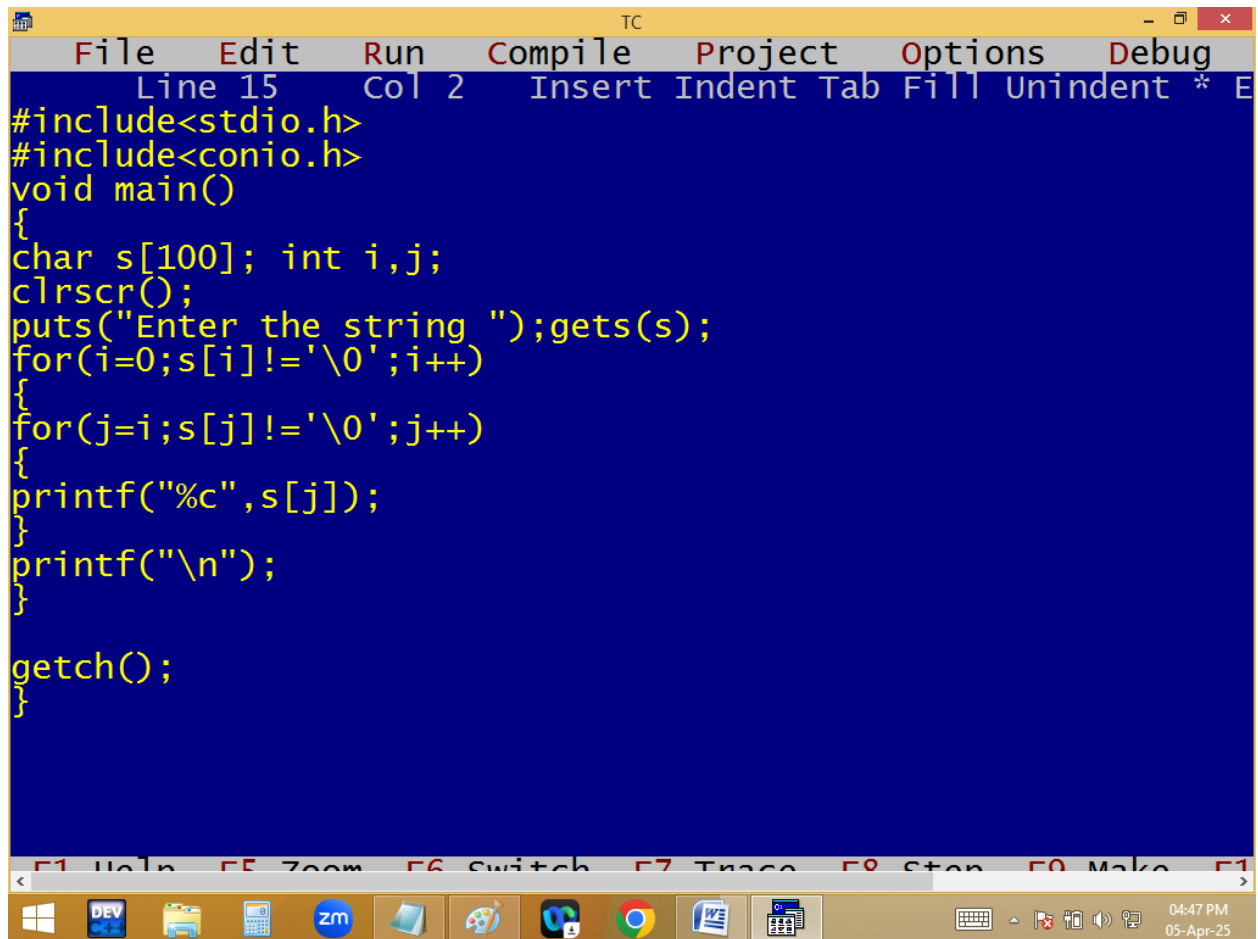
$65500 + 1 * 1 = 65501$ to $\backslash 0 \implies bcd$

$65500 + 2 * 1 = 65502$ to $\backslash 0 \implies cd$

$65500 + 3 * 1 = 65503$ to $\backslash 0 \implies d$

$65500 + 4 * 1 = 65504 = \backslash 0 \neq \backslash 0 \implies \text{false}$

Method2:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 15", "Col 2", and "Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i,j;
clrscr();
puts("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++)
{
for(j=i;s[j]!='\0';j++)
{
printf("%c",s[j]);
}
printf("\n");
}

getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "04:47 PM" and the date "05-Apr-25".

```

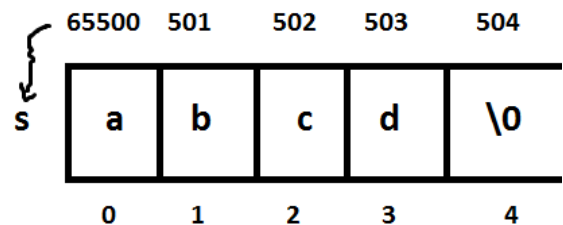
Enter the string
Kishore
Kishore
ishore
shore
hore
ore
re
e

```

```

for( i=0 ; s[i]!='\0'; i++ ) ✓
{
for( j=i; s[j]!='\0';j++)
{
p(s[j]);
}
}
p("\n");
}

```



$\frac{i}{0}$ $\frac{j}{0}$ to $\frac{j}{\text{to } \backslash 0}$ - abcd
1 1 // - bcd
2 2 // - cd
3 3 // - d

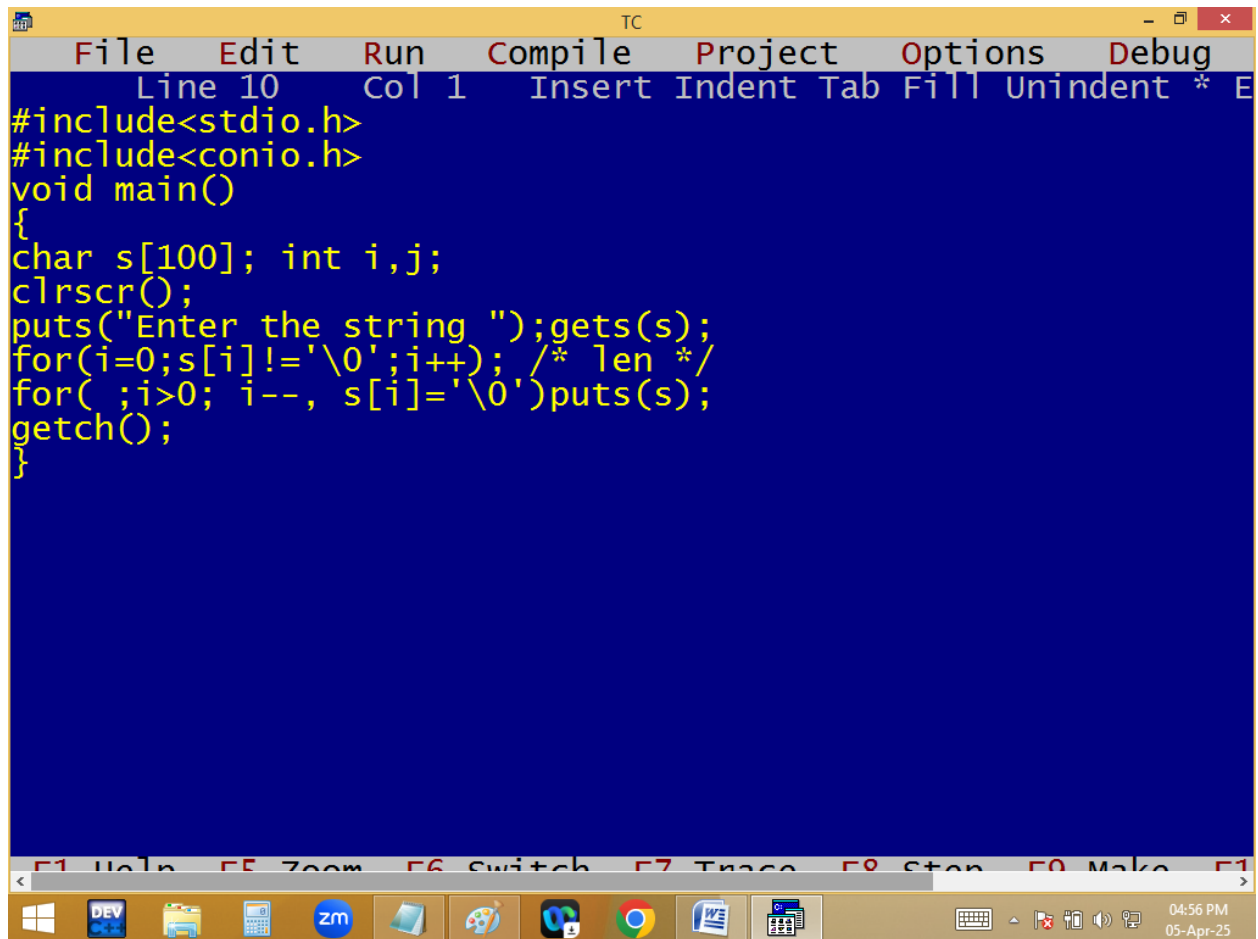
Given string → alia

Alia

Ali

Al

a



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 10", "Col 1", and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i,j;
clrscr();
puts("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++); /* len */
for( ;i>0; i--, s[i]='\0')puts(s);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "04:56 PM" and "05-Apr-25".

```
TC
Enter the string
Madhubala
Madhubala
Madhubal
Madhuba
Madhub
Madhu
Madh
Mad
Ma
M
```

for(i=0; s[i]!='\0';i++); i=4

for(; i> 0; i--, s[i]='\0') puts(s);



4
3
2
1
0

65500 to \0 = abcd

65500 to \0 = abc

65500 to \0 = ab

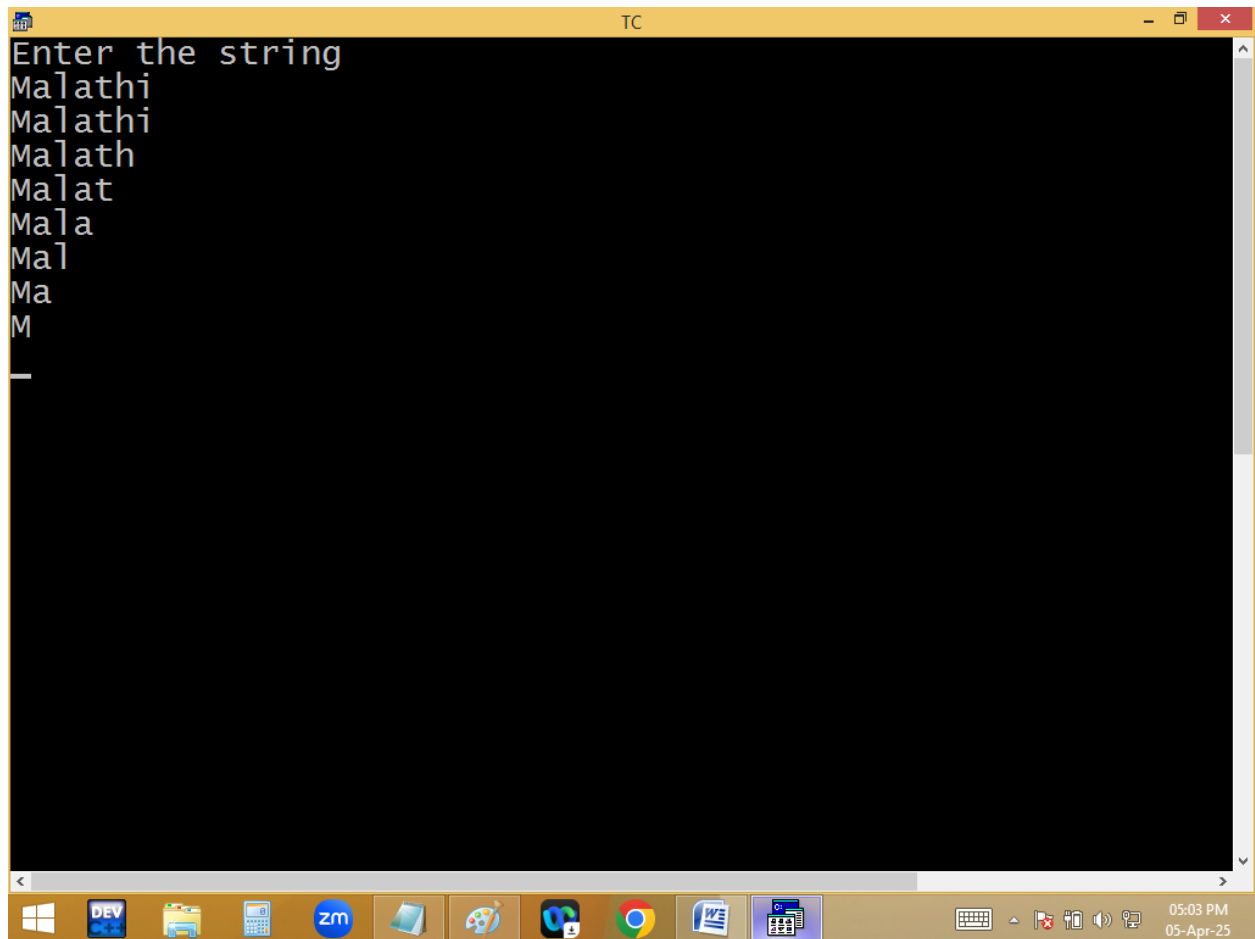
65500 to \0 = a

Method2:

```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 17 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i,j;
clrscr();
puts("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++); /* len */
for( i--; i>=0; i-- )
{
for( j=0;j<=i; j++)
{
printf("%c",s[j]);
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:03 PM
05-Apr-25



```
for( i=0; s[i]!='\0';i++); i=4
```

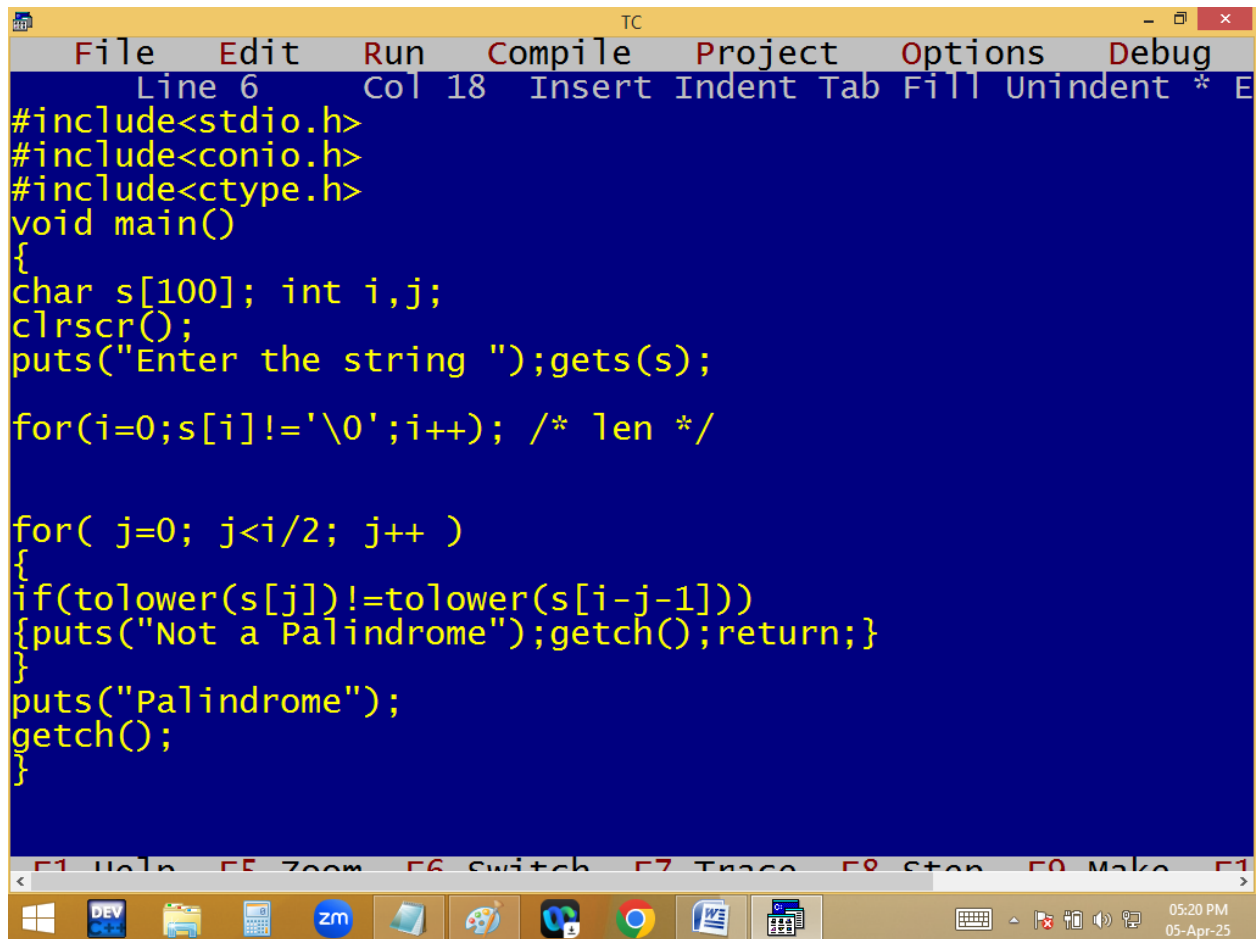
```
for( i--; i>=0; i-- )
{
    for( j=0; j<=i; j++ )
    {
        p(s[ j]);
    }
    p("\n");
}
```



$\downarrow$ $\downarrow$
 $0 - 4 \rightarrow abcd$
 $0 - 3 \rightarrow abc$
 $0 - 2 \rightarrow ab$
 $0 - 1 \rightarrow a$
 $0 - 0 \rightarrow$

Finding palindrome or not using single string?

Eg: lilil, madam, Malayalam,...



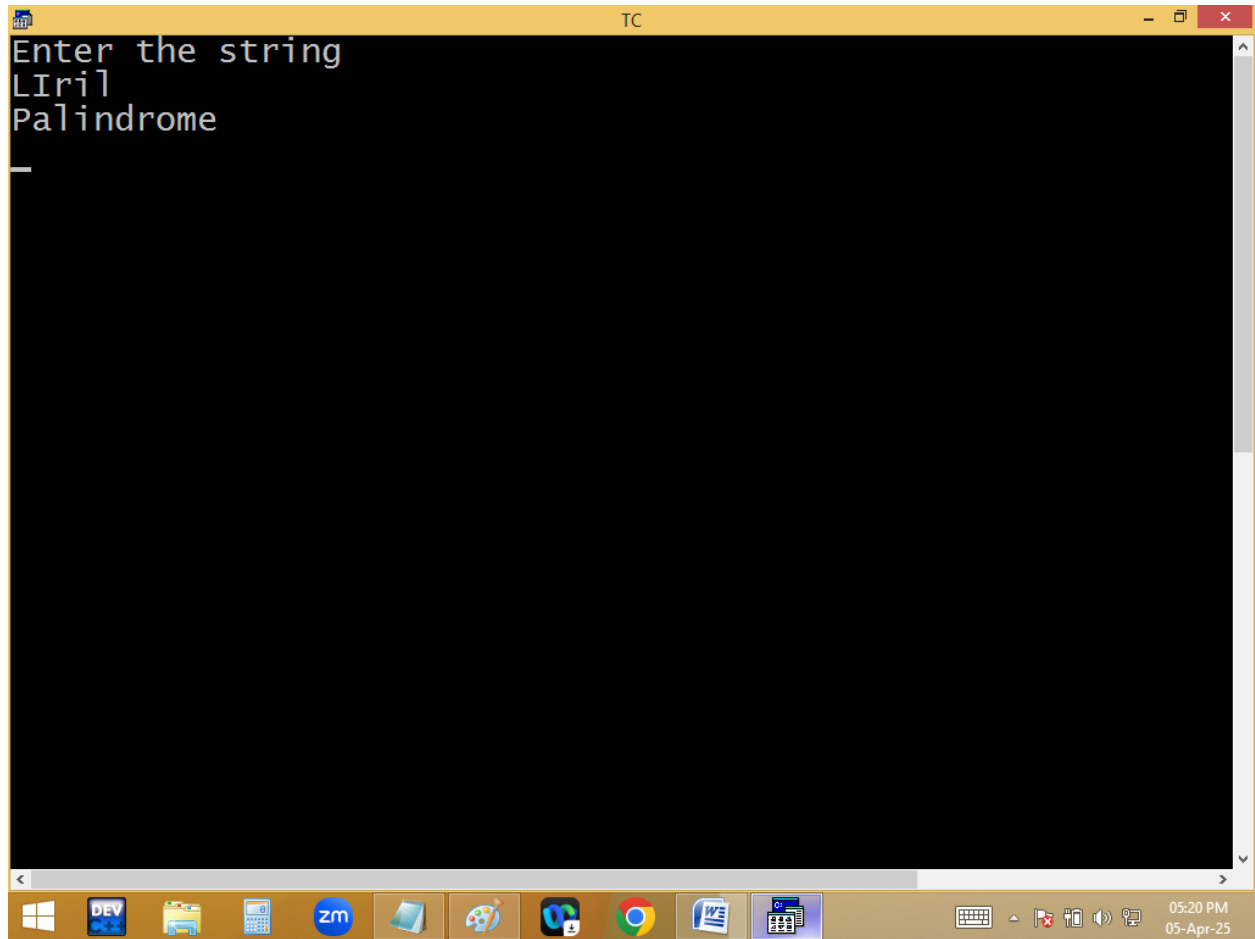
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 6" and "Col 18". The main editing area has a dark blue background with yellow text. The code is a C program to check if a string is a palindrome. It includes headers for stdio, conio, and ctype. The main function declares a character array 's' of size 100 and variables 'i' and 'j'. It prompts the user to enter a string using 'gets(s)'. Then, it uses a nested loop to compare characters from the start and end of the string, converting them to lowercase using 'tolower'. If a mismatch is found, it prints "Not a Palindrome" and returns. Otherwise, it prints "Palindrome". The bottom status bar shows function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom shows various application icons and the system clock indicating 05:20 PM on 05-Apr-25.

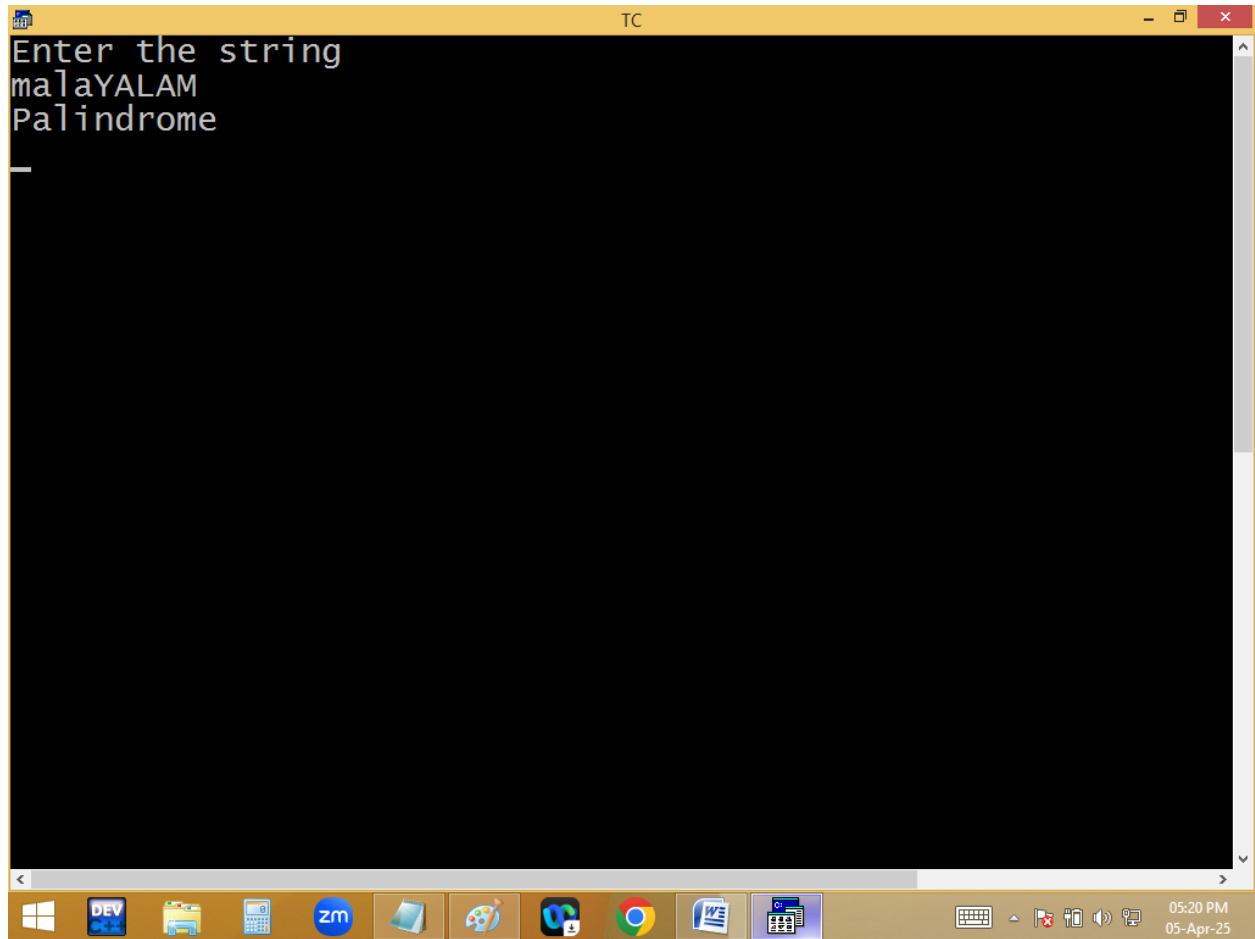
```
File Edit Run Compile Project Options Debug
Line 6 Col 18 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char s[100]; int i,j;
clrscr();
puts("Enter the string ");gets(s);

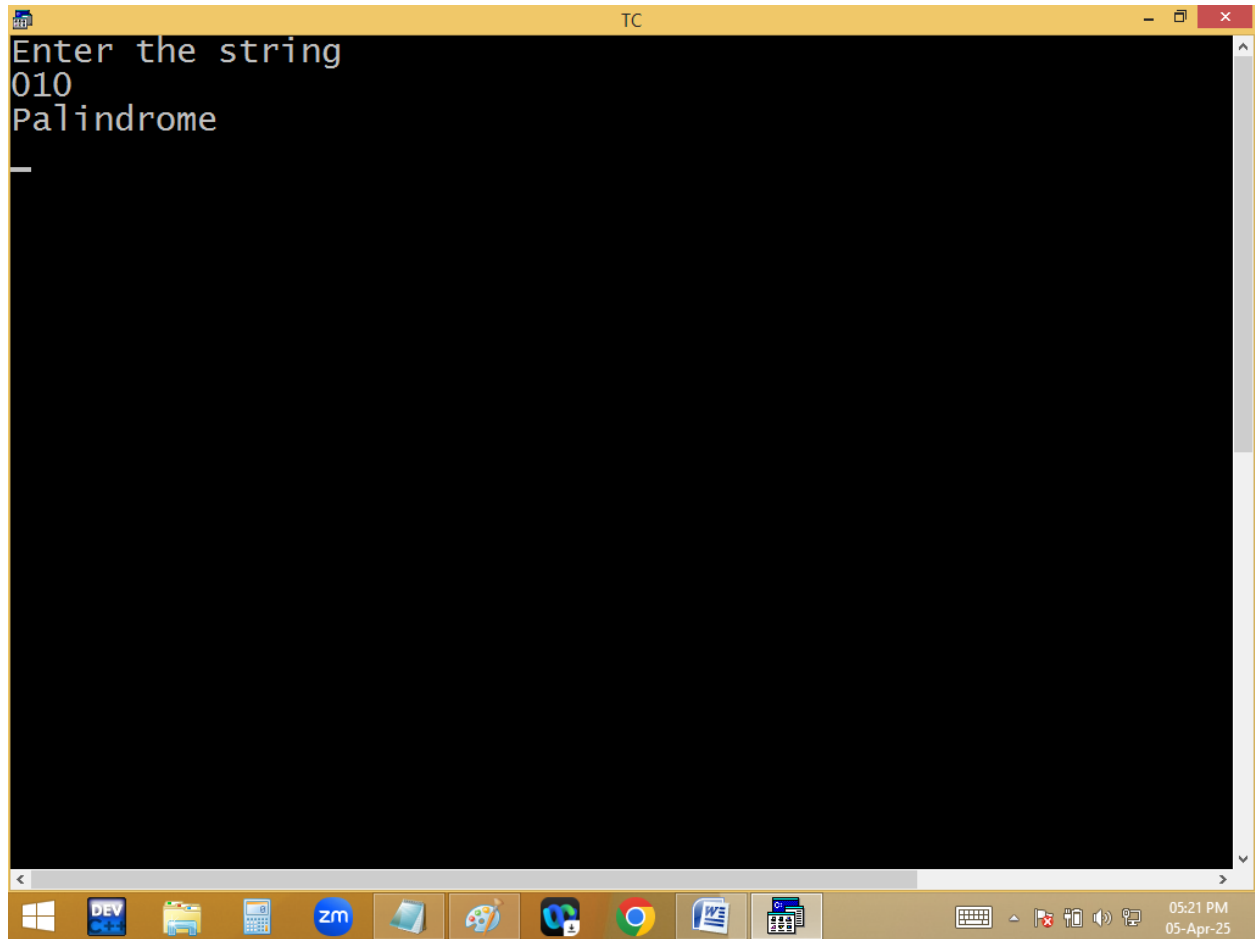
for(i=0;s[i]!='\0';i++); /* len */

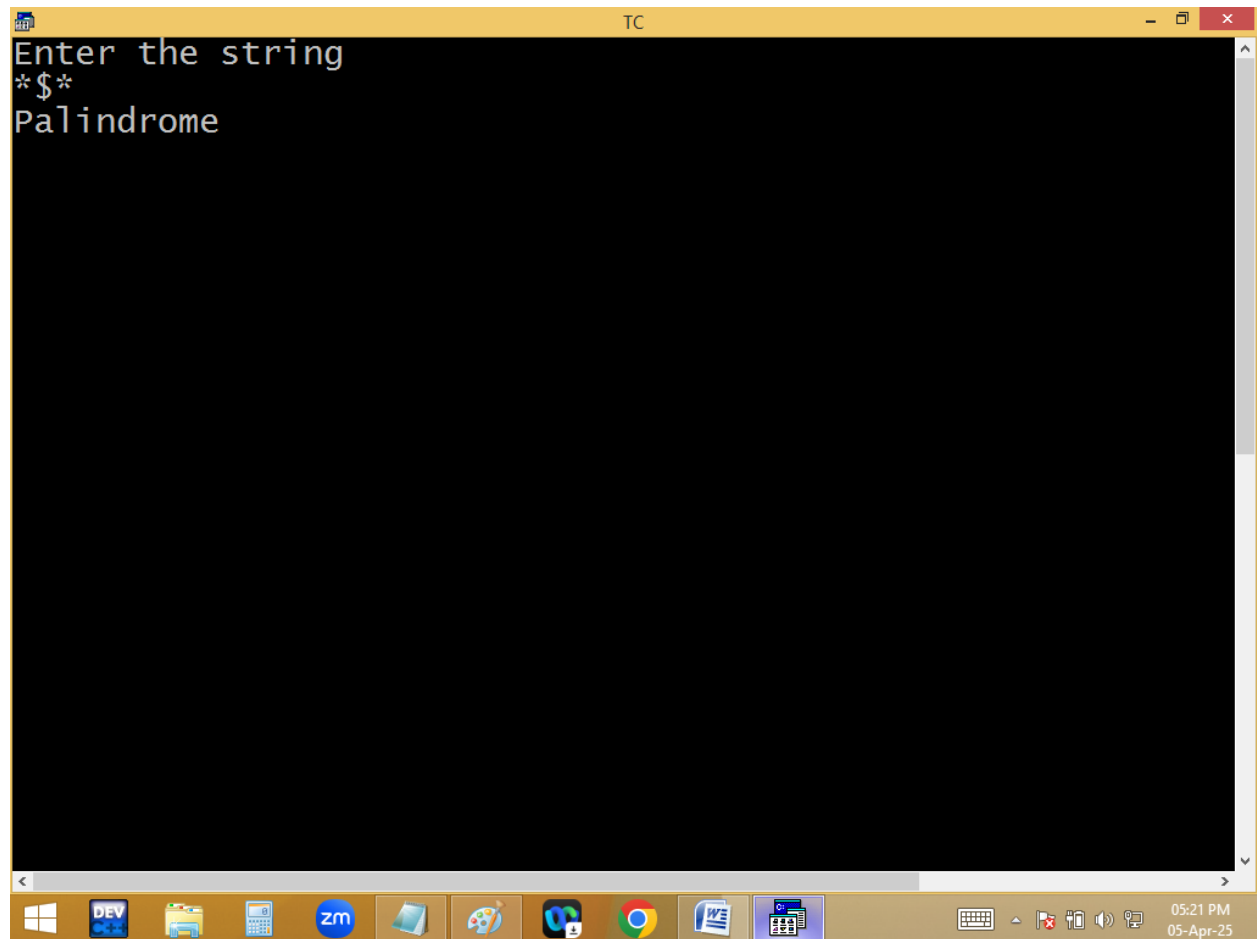
for( j=0; j<i/2; j++ )
{
if(tolower(s[j])!=tolower(s[i-j-1]))
{puts("Not a Palindrome");getch();return;}
}
puts("Palindrome");
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10









```
TC
Enter the string
AMMU
Not a Palindrome
```

for(i=0; s[i]!='\0';i++); i=4 ✓

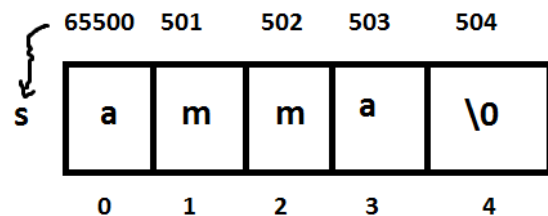
for(j=0; j<i/2; j++)

{
if(a[i-j-1] != a[j])

{
p("Not a palindrome"); return;

}
}

P(palindrome);



$$\begin{array}{r} j \\ 0 \\ 1 \\ 3 \end{array} \quad \begin{array}{r} i-j-1 \\ 4-0-1=3 \\ 4-1-1=2 \end{array}$$

Using 2 strings:

```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 32 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char s1[100],s2[100]; int i,j; clrscr();
puts("Enter the string ");gets(s1);

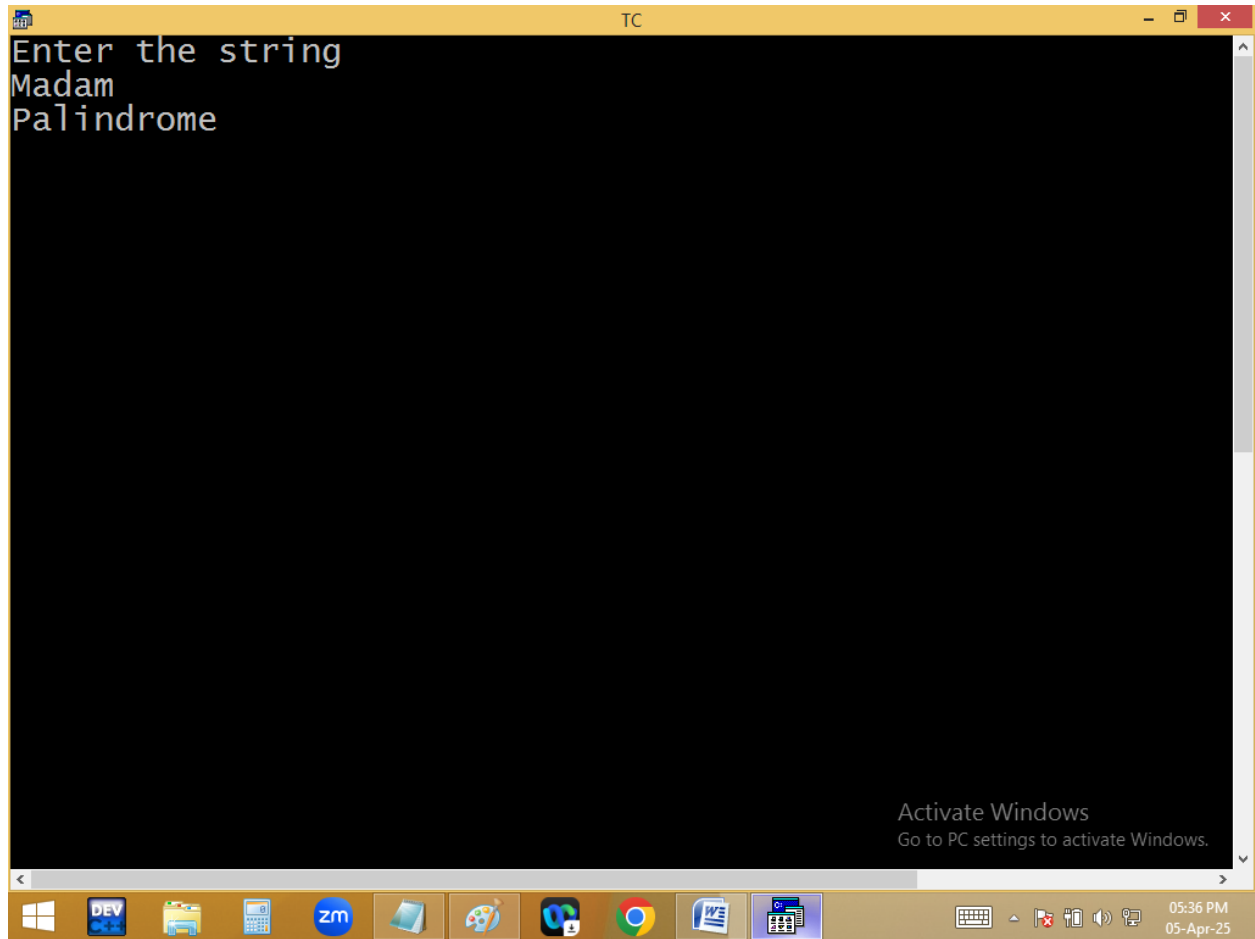
for(i=0;s1[i]!='\0';i++); /* len */
for( s2[i--]='\0',j=0; i>=0; i--,j++ )
{
s2[j]=s1[i];/*str rev + copy */
}

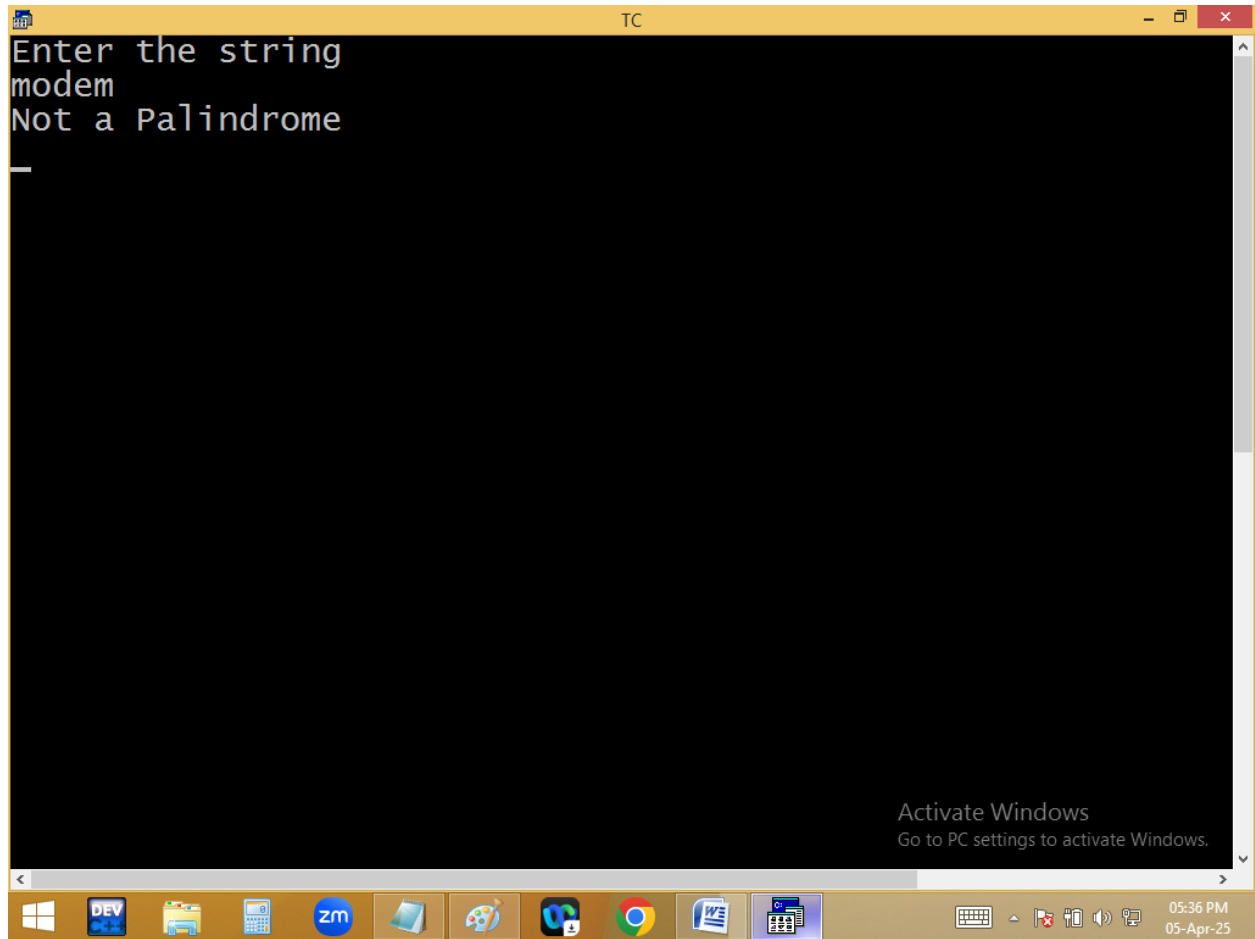
for(i=0; s1[i]!='\0';i++)
{
if(tolower(s1[i])!=tolower(s2[i])) /* strcmp */
{puts("Not a Palindrome");getch();return;}
}
puts("Palindrome");
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:36 PM
05-Apr-25





```

Enter the string
121
Palindrome

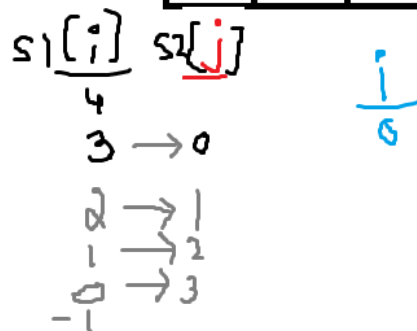
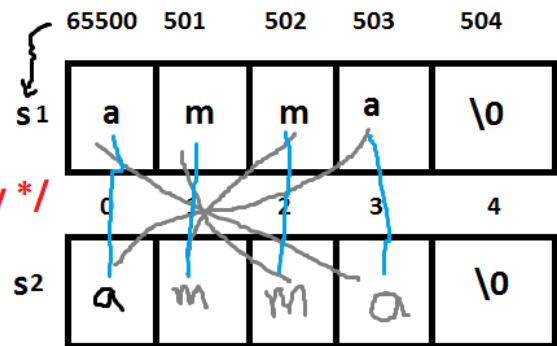
```

Activate Windows
Go to PC settings to activate Windows.

for(i=0; s[i]!='\0';i++); i=4 ✓ **strlen**

for(s2[i--]='\0', j=0; i>=0;i--,j++)
s2[j]=s1[i]; **/* rev+copy */**

for(i=0;s1[i]!='\0';i++)
{
if(s1[i]!=s2[i]) **/* strcmp */**
{
p(not);return;
}
}
p(palin);



Home work?

1. String concatenation?

Kishore + naidu = kishorenaidu

2. Kabhi Kushi Kabhi Ghum → K K K G – Abbreviation

Ranam Raudram Rudhiram → R R R

Kolar Gold Fields → K G F

3. I Miss You Jaanu → 4 words

4. silent , listen – anagram strings